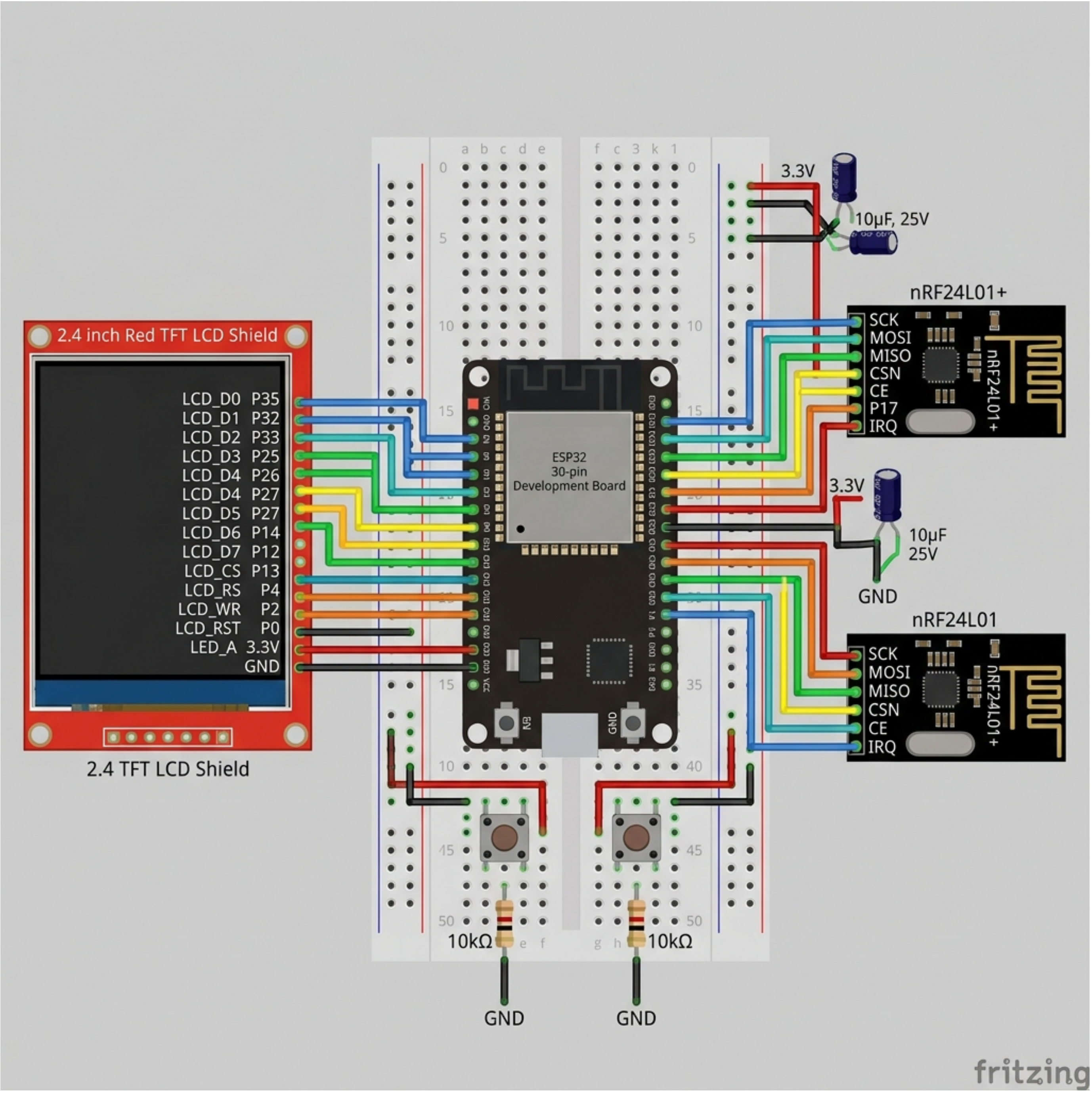
ESP32 BT JAMMER & WI-FI SECURITY SUITE

Master Hardware Pinout, RF Transceiver & Wiring Diagram Guide

LEGAL & EDUCATIONAL DISCLAIMER:

This device design and documentation are strictly for educational and authorized penetration testing in controlled environments. Operating RF jammers or packet injection without authorization is illegal.

1. Full Master Hardware Breadboard Wiring Diagram



2. ESP32 Pin Allocation Overview

ESP32 30-PIN DEV BOARD LAYOUT

Radio 1 CE	**GPIO 22**	**GPIO 21**	Radio 1 CSN
Radio 2 CE	**GPIO 16**	**GPIO 15**	Radio 2 CSN
Touch T_CS	**GPIO 27**	**GPIO 4**	TFT DC / RS
TFT Backlight	**GPIO 32**	**GPIO 5**	TFT CS
TFT Reset	**GPIO 33**	**GPIO 18**	SPI SCK (Clock)
Physical ON Btn	**GPIO 12**	**GPIO 19**	SPI MISO
Physical OFF Btn	**GPIO 14**	**GPIO 23**	SPI MOSI
Power Rail	**3.3V**	**GND**	Ground Rail

3. Hardware Interfaces (VSPI & ESP32 Wi-Fi Radio)

Signal / Interface	ESP32 Pin	Connected Modules / Mode	Type
SCK (Clock)	**GPIO 18**	Radio 1, Radio 2, TFT SCK, Touch T_CLK	Shared SPI
MISO (Master In)	**GPIO 19**	Radio 1, Radio 2, TFT SDO, Touch T_DO	Shared SPI
MOSI (Master Out)	**GPIO 23**	Radio 1, Radio 2, TFT SDI, Touch T_DIN	Shared SPI
Wi-Fi 802.11 b/g/n	**Internal RF**	AP Scanning, Promiscuous Sniffer, Web Portal	Internal RF

4. Radio Transceiver Modules (NRF24L01)

Radio #1 (Lower Channels 2402-2440 MHz):

NRF24 #1 Pin	ESP32 Pin	Function	Signal Category
VCC / GND	**3.3V / GND**	Power Rail (+3.3V Max)	Power / Ground
CE	**GPIO 22**	Chip Enable	Dedicated Control
CSN	**GPIO 21**	SPI Chip Select	Dedicated SPI CS
SCK / MOSI / MISO	**GPIO 18 / 23 / 19**	Hardware SPI Bus	Shared SPI

Radio #2 (Upper Channels 2441-2480 MHz):

NRF24 #2 Pin	ESP32 Pin	Function	Signal Category
VCC / GND	**3.3V / GND**	Power Rail (+3.3V Max)	Power / Ground
CE	**GPIO 16**	Chip Enable	Dedicated Control
CSN	**GPIO 15**	SPI Chip Select	Dedicated SPI CS
SCK / MOSI / MISO	**GPIO 18 / 23 / 19**	Hardware SPI Bus	Shared SPI

5. 2.4" TFT LCD Shield (8-Bit Parallel Bus)

Shield Pin	ESP32 Pin	Function	Signal Category
5V / 3V3 / GND	**5V / 3.3V / GND**	Display Power & Ground	Power / Ground
LCD_RST	**GPIO 33**	Hardware Reset	Dedicated Control
LCD_CS	**GPIO 5**	Chip Select	Dedicated Control
LCD_RS	**GPIO 4**	Register / Command Select	Dedicated Control
LCD_WR	**GPIO 2**	Write Enable	Dedicated Control
LCD_RD	**3.3V (Tie HIGH)**	Read Enable (Not Used)	Power
LCD_D0	**GPIO 12**	Data Bit 0	8-Bit Parallel
LCD_D1	**GPIO 13**	Data Bit 1	8-Bit Parallel
LCD_D2	**GPIO 26**	Data Bit 2	8-Bit Parallel
LCD_D3	**GPIO 25**	Data Bit 3	8-Bit Parallel
LCD_D4	**GPIO 17**	Data Bit 4	8-Bit Parallel
LCD_D5	**GPIO 27**	Data Bit 5	8-Bit Parallel
LCD_D6	**GPIO 14**	Data Bit 6	8-Bit Parallel
LCD_D7	**GPIO 32**	Data Bit 7	8-Bit Parallel

6. Hardware Pushbuttons & System Controls

Button / LED	ESP32 Pin	Wiring Connection	Mode
TURN ON Button	**GPIO 12**	One leg to GPIO 12, other leg to GND	INPUT_PULLUP
TURN OFF Button	**GPIO 14**	One leg to GPIO 14, other leg to GND	INPUT_PULLUP
BOOT Button	**GPIO 0**	Built-in BOOT pushbutton on ESP32 board	Internal
Status LED	**GPIO 2**	Built-in LED on ESP32 board	OUTPUT

7. Power Supply & Capacitor Recommendations

1. Built-in PCB Antenna Modules: Draw ~12mA each. Can be powered directly from ESP32 3.3V pin without capacitors.
2. High-Power PA+LNA Modules: Draw ~115mA peak each. Add a 10uF-47uF capacitor across VCC & GND of each module to prevent brownouts.